DESSERTOPIA

Database Relationship Analysis Using Joins

Introduction :

Database Relationship Analysis using Joins is used to retrieve related information from multiple tables in the E-Commerce Database. In this project, SQL JOIN operations are used to connect the CUSTOMER, ORDERS, ORDER_DETAILS, PRODUCT, and PAYMENT tables through their common columns, such as CUSTOMER_ID, ORDER_ID, and PRODUCT_ID.

Using these joins, the queries identify all products purchased by a customer, calculate the total amount spent by each customer, and find the number of orders placed by each customer. They can also retrieve order details along with payment status. These queries help analyze customer purchases, order history, and payment information efficiently and provide a better understanding of the relationships between the tables in the database.

Combine Related Tables Using Joins :

```
SQL> SELECT O.*,PY.* FROM ORDERS O JOIN PAYMENT PY ON O.ORDER_ID = PY.ORDER_ID;

ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS         PAYMENT_ID
---------- ---------- --------- ------------ -------------------- ----------
ORDER_ID   PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
O001       C001       01-OCT-26          550 Pending              PAY001
O001       01-OCT-26 UPI                  Success
               550

O002       C002       02-OCT-26          240 Shipped              PAY002
O002       02-OCT-26 Card                 Success
               240

ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS         PAYMENT_ID
---------- ---------- --------- ------------ -------------------- ----------
ORDER_ID   PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------

O003       C003       03-OCT-26          360 Delivered            PAY003
O003       03-OCT-26 Cash                 Failed
               360

O004       C004       04-OCT-26          300 Pending              PAY004
O004       04-OCT-26 Net Banking          Pending
```

```
ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS         PAYMENT_ID
---------- ---------- --------- ------------ -------------------- ----------
ORDER_ID   PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
               300

O005       C005       05-OCT-26          500 Shipped              PAY005
O005       05-OCT-26 UPI                  Success
               500

O006       C006       06-OCT-26          450 Delivered            PAY006

ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS         PAYMENT_ID
---------- ---------- --------- ------------ -------------------- ----------
ORDER_ID   PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
O006       06-OCT-26 Card                 Failed
               450

O007       C007       07-OCT-26          400 Pending              PAY007
O007       07-OCT-26 UPI                  Success
               400
```

Implementing Inner Join :

```
SQL> SELECT C.*,O.* FROM CUSTOMER C INNER JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_I FIRST_NAME                     EMAIL_ID
---------- ------------------------------ ------------------------------
PHONE_NUMBER
------------
ADDRESS
--------------------------------------------------------------------------------
CREATED_A LAST_NAME            ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT
--------- -------------------- ---------- ---------- --------- ------------
ORDER_STATUS
--------------------
C001       Kavya                          kavya@gmail.com
  9876543201
Chennai

CUSTOMER_I FIRST_NAME                     EMAIL_ID
---------- ------------------------------ ------------------------------
PHONE_NUMBER
------------
ADDRESS
--------------------------------------------------------------------------------
CREATED_A LAST_NAME            ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT
--------- -------------------- ---------- ---------- --------- ------------
ORDER_STATUS
--------------------
01-OCT-26 Dharshini            O001       C001       01-OCT-26          550
Pending
```

```
CUSTOMER_I FIRST_NAME                     EMAIL_ID
---------- ------------------------------ ------------------------------
PHONE_NUMBER
------------
ADDRESS
--------------------------------------------------------------------------------
CREATED_A LAST_NAME            ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT
--------- -------------------- ---------- ---------- --------- ------------
ORDER_STATUS
--------------------
C002       Roshini                        roshini@gmail.com
  9876543202
Chennai

CUSTOMER_I FIRST_NAME                     EMAIL_ID
---------- ------------------------------ ------------------------------
PHONE_NUMBER
------------
ADDRESS
--------------------------------------------------------------------------------
CREATED_A LAST_NAME            ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT
--------- -------------------- ---------- ---------- --------- ------------
ORDER_STATUS
--------------------
02-OCT-26 Priya                O002       C002       02-OCT-26          240
Shipped
```

```
CUSTOMER_I FIRST_NAME                     EMAIL_ID
---------- ------------------------------ ------------------------------
PHONE_NUMBER
------------
ADDRESS
--------------------------------------------------------------------------------
CREATED_A LAST_NAME            ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT
--------- -------------------- ---------- ---------- --------- ------------
ORDER_STATUS
--------------------
C003       Ananya                         ananya@gmail.com
  9876543203
Coimbatore

CUSTOMER_I FIRST_NAME                     EMAIL_ID
---------- ------------------------------ ------------------------------
PHONE_NUMBER
------------
ADDRESS
--------------------------------------------------------------------------------
CREATED_A LAST_NAME            ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT
--------- -------------------- ---------- ---------- --------- ------------
ORDER_STATUS
--------------------
03-OCT-26 Krishnan             O003       C003       03-OCT-26          360
Delivered
```

```
CUSTOMER_I FIRST_NAME                     EMAIL_ID
---------- ------------------------------ ------------------------------
PHONE_NUMBER
------------
ADDRESS
--------------------------------------------------------------------------------
CREATED_A LAST_NAME            ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT
--------- -------------------- ---------- ---------- --------- ------------
ORDER_STATUS
--------------------
C004       Priya                          priya@gmail.com
  9876543204
Madurai

CUSTOMER_I FIRST_NAME                     EMAIL_ID
---------- ------------------------------ ------------------------------
PHONE_NUMBER
------------
ADDRESS
--------------------------------------------------------------------------------
CREATED_A LAST_NAME            ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT
--------- -------------------- ---------- ---------- --------- ------------
ORDER_STATUS
--------------------
04-OCT-26 Sharma               O004       C004       04-OCT-26          300
Pending
```

```
CUSTOMER_I FIRST_NAME                     EMAIL_ID
---------- ------------------------------ ------------------------------
PHONE_NUMBER
------------
ADDRESS
--------------------------------------------------------------------------------
CREATED_A LAST_NAME            ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT
--------- -------------------- ---------- ---------- --------- ------------
ORDER_STATUS
--------------------
C005       Harini                         harini@gmail.com
  9876543205
Bengaluru

CUSTOMER_I FIRST_NAME                     EMAIL_ID
---------- ------------------------------ ------------------------------
PHONE_NUMBER
------------
ADDRESS
--------------------------------------------------------------------------------
CREATED_A LAST_NAME            ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT
--------- -------------------- ---------- ---------- --------- ------------
ORDER_STATUS
--------------------
05-OCT-26 Kumar                O005       C005       05-OCT-26          500
Shipped
```

```
CUSTOMER_I FIRST_NAME                     EMAIL_ID
---------- ------------------------------ ------------------------------
PHONE_NUMBER
------------
ADDRESS
--------------------------------------------------------------------------------
CREATED_A LAST_NAME            ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT
--------- -------------------- ---------- ---------- --------- ------------
ORDER_STATUS
--------------------
C006       Divya                          divya@gmail.com
  9876543206
Chennai

CUSTOMER_I FIRST_NAME                     EMAIL_ID
---------- ------------------------------ ------------------------------
PHONE_NUMBER
------------
ADDRESS
--------------------------------------------------------------------------------
CREATED_A LAST_NAME            ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT
--------- -------------------- ---------- ---------- --------- ------------
ORDER_STATUS
--------------------
06-OCT-26 Raj                  O006       C006       06-OCT-26          450
Delivered
```

```
CUSTOMER_I FIRST_NAME                    EMAIL_ID
---------- ------------------------------ ------------------------------
PHONE_NUMBER
------------
ADDRESS
--------------------------------------------------------------------------------
CREATED_A LAST_NAME            ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT
--------- -------------------- ---------- ---------- --------- ------------
ORDER_STATUS
--------------------
C007       Sneha                         sneha@gmail.com
  9876543207
Coimbatore

CUSTOMER_I FIRST_NAME                    EMAIL_ID
---------- ------------------------------ ------------------------------
PHONE_NUMBER
------------
ADDRESS
--------------------------------------------------------------------------------
CREATED_A LAST_NAME            ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT
--------- -------------------- ---------- ---------- --------- ------------
ORDER_STATUS
--------------------
07-OCT-26 Mohan                O007       C007       07-OCT-26         400
Pending


7 rows selected.
```

Order Details With Payment Information :

```
SQL> SELECT O.*,PY.* FROM ORDERS O JOIN PAYMENT PY ON O.ORDER_ID = PY.ORDER_ID;

ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS          PAYMENT_ID
---------- ---------- --------- ------------ --------------------- ----------
ORDER_ID   PAYMENT_D PAYMENT_MODE          PAYMENT_STATUS
---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
O0001      C001       01-OCT-26          550 Pending               PAY001
O0001      01-OCT-26 UPI                   Success
             550

O0002      C002       02-OCT-26          240 Shipped               PAY002
O0002      02-OCT-26 Card                  Success
             240

ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS          PAYMENT_ID
---------- ---------- --------- ------------ --------------------- ----------
ORDER_ID   PAYMENT_D PAYMENT_MODE          PAYMENT_STATUS
---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------

O0003      C003       03-OCT-26          360 Delivered             PAY003
O0003      03-OCT-26 Cash                  Failed
             360

O0004      C004       04-OCT-26          300 Pending               PAY004
O0004      04-OCT-26 Net Banking           Pending
```

```
ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS          PAYMENT_ID
---------- ---------- --------- ------------ --------------------- ----------
ORDER_ID   PAYMENT_D PAYMENT_MODE          PAYMENT_STATUS
---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
             300

O0005      C005       05-OCT-26          500 Shipped               PAY005
O0005      05-OCT-26 UPI                   Success
             500

O0006      C006       06-OCT-26          450 Delivered             PAY006

ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS          PAYMENT_ID
---------- ---------- --------- ------------ --------------------- ----------
ORDER_ID   PAYMENT_D PAYMENT_MODE          PAYMENT_STATUS
---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
O0006      06-OCT-26 Card                  Failed
             450

O0007      C007       07-OCT-26          400 Pending               PAY007
O0007      07-OCT-26 UPI                   Success
             400
```

Implement RIGHT JOIN :

Displaying All Orders With Customer Information :

```
SQL> SELECT O.* , C.*
  2  FROM CUSTOMER C RIGHT JOIN ORDERS O ON O.CUSTOMER_ID = C.CUSTOMER_ID;

ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS         CUSTOMER_I
---------- ---------- --------- ------------ -------------------- ----------
FIRST_NAME                     EMAIL_ID                       PHONE_NUMBER
------------------------------ ------------------------------ ------------
ADDRESS
--------------------------------------------------------------------------------
CREATED_A LAST_NAME
--------- --------------------
O002       C002       02-OCT-26          240 Shipped              C002
Roshini                        roshini@gmail.com              9876543202
Chennai
02-OCT-26 Priya


ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS         CUSTOMER_I
---------- ---------- --------- ------------ -------------------- ----------
FIRST_NAME                     EMAIL_ID                       PHONE_NUMBER
------------------------------ ------------------------------ ------------
ADDRESS
--------------------------------------------------------------------------------
CREATED_A LAST_NAME
--------- --------------------
O003       C003       03-OCT-26          360 Delivered            C003
Ananya                         ananya@gmail.com               9876543203
Coimbatore
03-OCT-26 Krishnan
```

```
ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS         CUSTOMER_I
---------- ---------- --------- ------------ -------------------- ----------
FIRST_NAME                     EMAIL_ID                       PHONE_NUMBER
------------------------------ ------------------------------ ------------
ADDRESS
--------------------------------------------------------------------------------
CREATED_A LAST_NAME
--------- --------------------
O004       C004       04-OCT-26          300 Pending              C004
Priya                          priya@gmail.com                9876543204
Madurai
04-OCT-26 Sharma


ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS         CUSTOMER_I
---------- ---------- --------- ------------ -------------------- ----------
FIRST_NAME                     EMAIL_ID                       PHONE_NUMBER
------------------------------ ------------------------------ ------------
ADDRESS
--------------------------------------------------------------------------------
CREATED_A LAST_NAME
--------- --------------------
O005       C005       05-OCT-26          500 Shipped              C005
Harini                         harini@gmail.com               9876543205
Bengaluru
05-OCT-26 Kumar
```

```
ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS         CUSTOMER_I
---------- ---------- --------- ------------ -------------------- ----------
FIRST_NAME                     EMAIL_ID                       PHONE_NUMBER
------------------------------ ------------------------------ ------------
ADDRESS
--------------------------------------------------------------------------------
CREATED_A LAST_NAME
--------- --------------------
O006       C006       06-OCT-26          450 Delivered            C006
Divya                          divya@gmail.com                  9876543206
Chennai
06-OCT-26 Raj


ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS         CUSTOMER_I
---------- ---------- --------- ------------ -------------------- ----------
FIRST_NAME                     EMAIL_ID                       PHONE_NUMBER
------------------------------ ------------------------------ ------------
ADDRESS
--------------------------------------------------------------------------------
CREATED_A LAST_NAME
--------- --------------------
O007       C007       07-OCT-26          400 Pending              C007
Sneha                          sneha@gmail.com                  9876543207
Coimbatore
07-OCT-26 Mohan
```

```
ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS         CUSTOMER_I
---------- ---------- --------- ------------ -------------------- ----------
FIRST_NAME                     EMAIL_ID                       PHONE_NUMBER
------------------------------ ------------------------------ ------------
ADDRESS
--------------------------------------------------------------------------------
CREATED_A LAST_NAME
--------- --------------------
O001       C001       01-OCT-26          550 Pending              C001
Kavya                          kavya@gmail.com                  9876543201
Chennai
01-OCT-26 Dharshini


7 rows selected.
```

Display all payment records with order details :

```
SQL> SELECT O.*,PY.* FROM ORDERS O RIGHT JOIN PAYMENT PY ON O.ORDER_ID = PY.ORDER_ID;

ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS         PAYMENT_ID
---------- ---------- --------- ------------ -------------------- ----------
ORDER_ID   PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
O001       C001       01-OCT-26          550 Pending              PAY001
O001       01-OCT-26 UPI                  Success
              550

O002       C002       02-OCT-26          240 Shipped              PAY002
O002       02-OCT-26 Card                 Success
              240

ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS         PAYMENT_ID
---------- ---------- --------- ------------ -------------------- ----------
ORDER_ID   PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------

O003       C003       03-OCT-26          360 Delivered            PAY003
O003       03-OCT-26 Cash                 Failed
              360

O004       C004       04-OCT-26          300 Pending              PAY004
O004       04-OCT-26 Net Banking          Pending
```

```
ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS         PAYMENT_ID
---------- ---------- --------- ------------ -------------------- ----------
ORDER_ID   PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
               300

O005       C005       05-OCT-26          500 Shipped              PAY005
O005       05-OCT-26 UPI                   Success
               500

O006       C006       06-OCT-26          450 Delivered            PAY006

ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS         PAYMENT_ID
---------- ---------- --------- ------------ -------------------- ----------
ORDER_ID   PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
O006       06-OCT-26 Card                  Failed
               450

O007       C007       07-OCT-26          400 Pending              PAY007
O007       07-OCT-26 UPI                   Success
               400


7 rows selected.
```

Retrieve Complete Order Details :

```
SQL> SELECT C.FIRST_NAME,C.LAST_NAME,P.PRODUCT_NAME,OD.QUANTITY,O.TOTAL_AMOUNT,PY.PAYMENT_STATUS,O.ORDER_DATE FROM CUSTOMER C INNER J
OIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID INNER JOIN PAYMENT PY ON O.ORDER_ID = PY.ORDER_ID INNER JOIN ORDER_DETAILS OD ON O.ORDE
R_ID = OD.ORDER_ID INNER JOIN PRODUCT P ON OD.PRODUCT_ID = P.PRODUCT_ID WHERE UPPER(PY.PAYMENT_STATUS) = 'SUCCESS';

FIRST_NAME                     LAST_NAME            PRODUCT_NAME
------------------------------ -------------------- --------------------
  QUANTITY TOTAL_AMOUNT PAYMENT_STATUS       ORDER_DAT
---------- ------------ -------------------- ---------
Kavya                          Dharshini            CHOCOLATE CAKE
         1          550 Success              01-OCT-26

Roshini                        Priya                VANILLA CUPCAKE
         2          240 Success              02-OCT-26

Harini                         Kumar                GLAZED DONUT
         5          500 Success              05-OCT-26


FIRST_NAME                     LAST_NAME            PRODUCT_NAME
------------------------------ -------------------- --------------------
  QUANTITY TOTAL_AMOUNT PAYMENT_STATUS       ORDER_DAT
---------- ------------ -------------------- ---------
Sneha                          Mohan                RED VELVET ICE CREAM
         2          400 Success              07-OCT-26
```

Activate Windows

All products purchased by a customer :

```
SQL> SELECT P.PRODUCT_NAME,P.PRODUCT_ID FROM PRODUCT P INNER JOIN ORDER_DETAILS OD ON OD.PRODUCT_ID = P.PRODUCT_ID INNER JOIN ORDERS
O ON O.ORDER_ID = OD.ORDER_ID WHERE O.CUSTOMER_ID = 'C003';

PRODUCT_NAME         PRODUCT
-------------------- -------
CHOCO BROWNIE        P003

SQL> SELECT P.PRODUCT_NAME,P.PRODUCT_ID FROM PRODUCT P INNER JOIN ORDER_DETAILS OD ON OD.PRODUCT_ID = P.PRODUCT_ID INNER JOIN ORDERS
O ON O.ORDER_ID = OD.ORDER_ID WHERE O.CUSTOMER_ID = 'C005';

PRODUCT_NAME         PRODUCT
-------------------- -------
GLAZED DONUT         P005
```

Activate Windows

Total amount spent by each customer :

```
SQL> SELECT C.CUSTOMER_ID,C.FIRST_NAME,C.LAST_NAME, SUM(TOTAL_AMOUNT) AS TOTAL_SPENT FROM CUSTOMER C INNER JOIN ORDERS O ON C.CUSTOME
R_ID = O.CUSTOMER_ID GROUP BY C.CUSTOMER_ID,C.FIRST_NAME,C.LAST_NAME;

CUSTOMER_I FIRST_NAME                    LAST_NAME            TOTAL_SPENT
---------- ----------------------------- -------------------- -----------
C005       Harini                        Kumar                        500
C007       Sneha                         Mohan                        400
C001       Kavya                         Dharshini                    550
C003       Ananya                        Krishnan                     360
C004       Priya                         Sharma                       300
C006       Divya                         Raj                          450
C002       Roshini                       Priya                        240

7 rows selected.
```

Activate Windows

Number of orders placed by each customer :

```
SQL> SELECT C.CUSTOMER_ID,C.FIRST_NAME,C.LAST_NAME, COUNT(O.ORDER_ID) AS TOTAL_ORDERS FROM CUSTOMER C INNER JOIN ORDERS O ON C.CUSTOM
ER_ID = O.CUSTOMER_ID GROUP BY C.CUSTOMER_ID,C.FIRST_NAME,C.LAST_NAME;

CUSTOMER_I FIRST_NAME                    LAST_NAME            TOTAL_ORDERS
---------- ----------------------------- -------------------- ------------
C005       Harini                        Kumar                           1
C007       Sneha                         Mohan                           1
C001       Kavya                         Dharshini                       1
C003       Ananya                        Krishnan                        1
C004       Priya                         Sharma                          1
C006       Divya                         Raj                             1
C002       Roshini                       Priya                           1

7 rows selected.
```

Activate Windows

Generate Multi-Table Business Reports :

Report 1:

Customer Order Report :

```
SQL> SELECT C.FIRST_NAME,C.LAST_NAME,O.ORDER_ID,O.ORDER_DATE,O.ORDER_STATUS FROM CUSTOMER C INNER JOIN ORDERS O ON C.CUSTOMER_ID= O.C
USTOMER_ID;

FIRST_NAME                    LAST_NAME            ORDER_ID   ORDER_DAT
----------------------------- -------------------- ---------- ---------
ORDER_STATUS
--------------------
Kavya                         Dharshini            O001       01-OCT-26
Pending

Roshini                       Priya                O002       02-OCT-26
Shipped

Ananya                        Krishnan             O003       03-OCT-26
Delivered


FIRST_NAME                    LAST_NAME            ORDER_ID   ORDER_DAT
----------------------------- -------------------- ---------- ---------
ORDER_STATUS
--------------------
Priya                         Sharma               O004       04-OCT-26
Pending

Harini                        Kumar                O005       05-OCT-26
Shipped

Divya                         Raj                  O006       06-OCT-26
Delivered
```

```
FIRST_NAME                    LAST_NAME            ORDER_ID   ORDER_DAT
----------------------------- -------------------- ---------- ---------
ORDER_STATUS
--------------------
Sneha                         Mohan                O007       07-OCT-26
Pending


7 rows selected.
```

Report 2:

Sales Report :

```
SQL> SELECT P.PRODUCT_NAME,SUM(OD.QUANTITY) AS QUANTITY_SOLD,SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE FROM PRODUCT P INNER JOIN O
RDER_DETAILS OD ON P.PRODUCT_ID = OD.PRODUCT_ID GROUP BY P.PRODUCT_ID,P.PRODUCT_NAME;

PRODUCT_NAME         QUANTITY_SOLD TOTAL_REVENUE
-------------------- ------------- -------------
CHOCOLATE CAKE                   1           550
VANILLA CUPCAKE                  2           240
CHOCO BROWNIE                    2           360
GLAZED DONUT                     5           500
BLUEBERRY CHEESECAKE             1           450
BUTTER COOKIES                   2           300
RED VELVET ICE CREAM             2           400

7 rows selected.
```

Report 3:

Payment Analysis Report :

```
SQL> SELECT PAYMENT_MODE,COUNT(*) AS TOTAL_TRANSACTIONS,SUM(CASE WHEN UPPER(PAYMENT_STATUS) = 'SUCCESS' THEN 1 ELSE 0 END) AS SUCCESS
FUL_PAYMENTS FROM PAYMENT GROUP BY PAYMENT_MODE;

PAYMENT_MODE         TOTAL_TRANSACTIONS SUCCESSFUL_PAYMENTS
-------------------- ------------------ -------------------
Net Banking                           1                   0
Card                                  2                   1
Cash                                  1                   0
UPI                                   3                   3
```

Activate Windows

Report 4:

Customer Purchase Analysis :

```
SQL> SELECT C.CUSTOMER_ID,C.FIRST_NAME,C.LAST_NAME,SUM(OD.QUANTITY) AS TOTAL_PRODUCTS_PURCHASED FROM CUSTOMER C INNER JOIN ORDERS O O
N C.CUSTOMER_ID = O.CUSTOMER_ID INNER JOIN ORDER_DETAILS OD ON O.ORDER_ID = OD.ORDER_ID GROUP BY C.CUSTOMER_ID,C.FIRST_NAME,C.LAST_NA
ME ORDER BY TOTAL_PRODUCTS_PURCHASED DESC;

CUSTOMER_I FIRST_NAME                    LAST_NAME
---------- ------------------------------ --------------------
TOTAL_PRODUCTS_PURCHASED
------------------------
C005       Harini                        Kumar
                      5

C007       Sneha                         Mohan
                      2

C003       Ananya                        Krishnan
                      2
```

```
CUSTOMER_I FIRST_NAME                    LAST_NAME
---------- ------------------------------ --------------------
TOTAL_PRODUCTS_PURCHASED
------------------------
C004       Priya                         Sharma
                      2

C002       Roshini                       Priya
                      2

C006       Divya                         Raj
                      1


CUSTOMER_I FIRST_NAME                    LAST_NAME
---------- ------------------------------ --------------------
TOTAL_PRODUCTS_PURCHASED
------------------------
C001       Kavya                         Dharshini
                      1


7 rows selected.
```

Implement LEFT JOIN :

```
SQL> SELECT C.CUSTOMER_ID,C.FIRST_NAME,C.LAST_NAME,O.ORDER_ID,O.ORDER_DATE FROM CUSTOMER C LEFT JOIN ORDERS O ON C.CUSTOMER_ID = O.CU
STOMER_ID;

CUSTOMER_I FIRST_NAME                    LAST_NAME           ORDER_ID
---------- ----------------------------- ------------------- ----------
ORDER_DAT
---------
C001       Kavya                         Dharshini           O001
01-OCT-26

C002       Roshini                       Priya               O002
02-OCT-26

C003       Ananya                        Krishnan            O003
03-OCT-26
```

```
CUSTOMER_I FIRST_NAME                    LAST_NAME           ORDER_ID
---------- ----------------------------- ------------------- ----------
ORDER_DAT
---------
C004       Priya                         Sharma              O004
04-OCT-26

C005       Harini                        Kumar               O005
05-OCT-26

C006       Divya                         Raj                 O006
06-OCT-26


CUSTOMER_I FIRST_NAME                    LAST_NAME           ORDER_ID
---------- ----------------------------- ------------------- ----------
ORDER_DAT
---------
C007       Sneha                         Mohan               O007
07-OCT-26


7 rows selected.
```